

[CO4]

The use of AI in biometric surveillance, particularly automated facial recognition (AFR) for crime-related tasks, is of utmost importance and must be accurate. If the report shows a false allegation against someone completely innocent, it could ruin someone's life. The humiliation and harassment that a person would face due to these false allegations could scar them for the rest of their life.

According to Buolamwini & Gebru (2018), error rates for facial analysis were 34.7% for dark-skinned women and 0.8% for lighter-skinned men. It is evident from the facts that facial recognition is more accurate for white-skinned men than for black or white-skinned women, turning the technology incompetent in its actual impact on the world (Balli, 2021). Even when there is no solid harm, pervasive surveillance can create gaps in the sense of security and generate anxiety among individuals who think that they are being monitored all the time.

Meanwhile, the long-term use of AI-enabled biometric surveillance being in place can be a bit concerning. Schep (n.d.) calls it "social cooling," meaning the gradual chilling of the free behavior of individuals who know that they are being watched and avoid being themselves so as not to get sanctioned. This can be detrimental in the long run and alter the relationship between the state and its citizens. There are also effects on the environment, as the infrastructure requires large-scale biometric surveillance, resulting in consuming a large amount of energy. The carbon footprints from these systems represent a growing environmental concern without any solid idea of tackling this issue.

[CO5]

In the field of AI, racial and gender bias in facial recognition systems is very common and evident. It created ethical violations through categorizing minorities and selective bias towards certain groups. Buolamwini & Gebru (2018) highlight that AFR is highly accurate for white men but lacks accuracy when it comes to dark/white-skinned women. The wrongful arrest of Nijer Parks shows that AI can be biased and that its algorithms are not 100% correct (Hill, 2020). The problem occurs when law enforcement officers treat these biased

outcomes as the standard. When standards get corrupted, it not only amplifies the racial inequalities but also crumbles the justice system.

Furthermore, biometric surveillance profoundly lacks transparency. People are unaware that their faces are used for training the AI models, which are being scanned and matched against law enforcement databases. Individuals in public spaces do not consent to having their biometrics used for purposes that they are unaware of. Reisman et al. (2018) point out that a person has the right to know when a computer algorithm makes a decision that affects that person's life. This should create a sense of understanding among the people that AI-enabled biometric surveillance still has transparency and can be trusted with. But the algorithm is too opaque to do so. Companies do not share all the information with the general public and guard their algorithm, making it difficult for the accused or their legal representative to know how the match was formed in the first place.

Moreover, the accountability of AI-enabled biometric surveillance is not that much on the good side. Whenever a wrongful accusation is proven, no one steps up and takes responsibility. A chain of blame games is seen, but the individual who suffered due to these false accusations gets little to no recourse. Some countries don't even have proper laws that deal with cases like these. A pattern can be seen where government agencies deploy many facial recognition systems without seeking accountability frameworks and legal authorization. This depicts the lack of responsibility as well as poor governance.

[CO6]

Deploying biometric surveillance in the real world carries significant and heavy legal risks on multiple levels. According to the EU's General Data Protection Regulation (GDPR), under Article 9, it is classified as personal data when biometric data is used for identifying a person, resulting in the processing of strict regulations of having explicit consent or specific legal authorization (European Parliament and Council of the EU, 2016). Without following these strict rules, an organization can face massive fines as well as deletion of the entire database.

Meanwhile, an engineer involved in designing these systems must have a clear conscience about what's right and what's wrong. They should have an ethical responsibility in creating models not for the sake of producing, but to ensure they have no bias and have transparency

and accountability. An engineer has many responsibilities and obligations, among which are the following:

- Training datasets are accurate and free of errors for the model to run properly.
- Conducting AI Impact Assessments (AI-IA) before deployment.
- Conducting bias auditing and dataset diversity to improve accuracy.
- Implementing clear and accessible notice for consent when surveillance is conducted.
- When an unreasonable risk to the public arises, an engineer must refuse to participate and report it accordingly.
- Designing systems that maintain privacy and data retention.
- Regular checking of the models to avoid bias.
- Holding accountability and transparency to the public when biometric surveillance is being done.

In conclusion, AI-enabled biometric surveillance systems are trained by humans. If bad datasets are given to it, the model will run its algorithm based on the data fed to it. It is the job of an engineer to know how to tackle this problem and make the model run so that no biases are projected in the outcomes. The sense of accountability and transparency is key to showing people they can trust the model and its results. As AI becomes more autonomous, those who design and deploy the system must own up to the mistakes the system creates, rather than pointing fingers at one another.

Reference

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